

Data types, structures and loops

Data Science Flipped Classroom - Module 6

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Quarto refresher

The three components of a .qmd file

- **YAML header:** metadata and options between `---` fences at the top
- **Markdown text:** headings, prose, lists, links
- **Code chunks:** executable R code between ````{r}` fences

Quarto runs the chunks, weaves the results into the text, and renders everything to HTML, PDF, or slides.

The YAML header

```
1 ---
2 title: "My analysis"
3 author: "Adriana Clavijo Daza"
4 date: "2026-06-04"
5 format: html
6 execute:
7   echo: true
8 ---
```

Indentation matters in YAML. Two spaces, never tabs.

Code chunk options

Options go inside the chunk, each on its own line starting with `#|`:

```
1 ```{r}
2 #| label: my-chunk
3 #| echo: true
4 #| warning: false
5
6 mean(c(1, 2, 3))
7 ```
```

The most useful chunk options

Option	Effect
<code>label</code>	names the chunk, useful for navigation and errors
<code>echo</code>	show the code itself
<code>eval</code>	run the code
<code>include</code>	show any output at all
<code>warning</code>	show warnings
<code>message</code>	show messages
<code>error</code>	render even if the chunk errors

Options for cross-references

Remember from module 5: two options turn a chunk's output into a **numbered, linkable** table or figure.

```
1 ```{r}
2 #| label: fig-height-mass
3 #| fig-cap: "Height vs mass of Star Wars characters"
4
5 starwars |>
6   ggplot(aes(height, mass)) +
7   geom_point()
8 ```
```

- `label` must start with `fig-` (figures) or `tbl-` (tables)
- Pair it with `fig-cap` or `tbl-cap`
- Refer to it in the text with `@fig-height-mass`